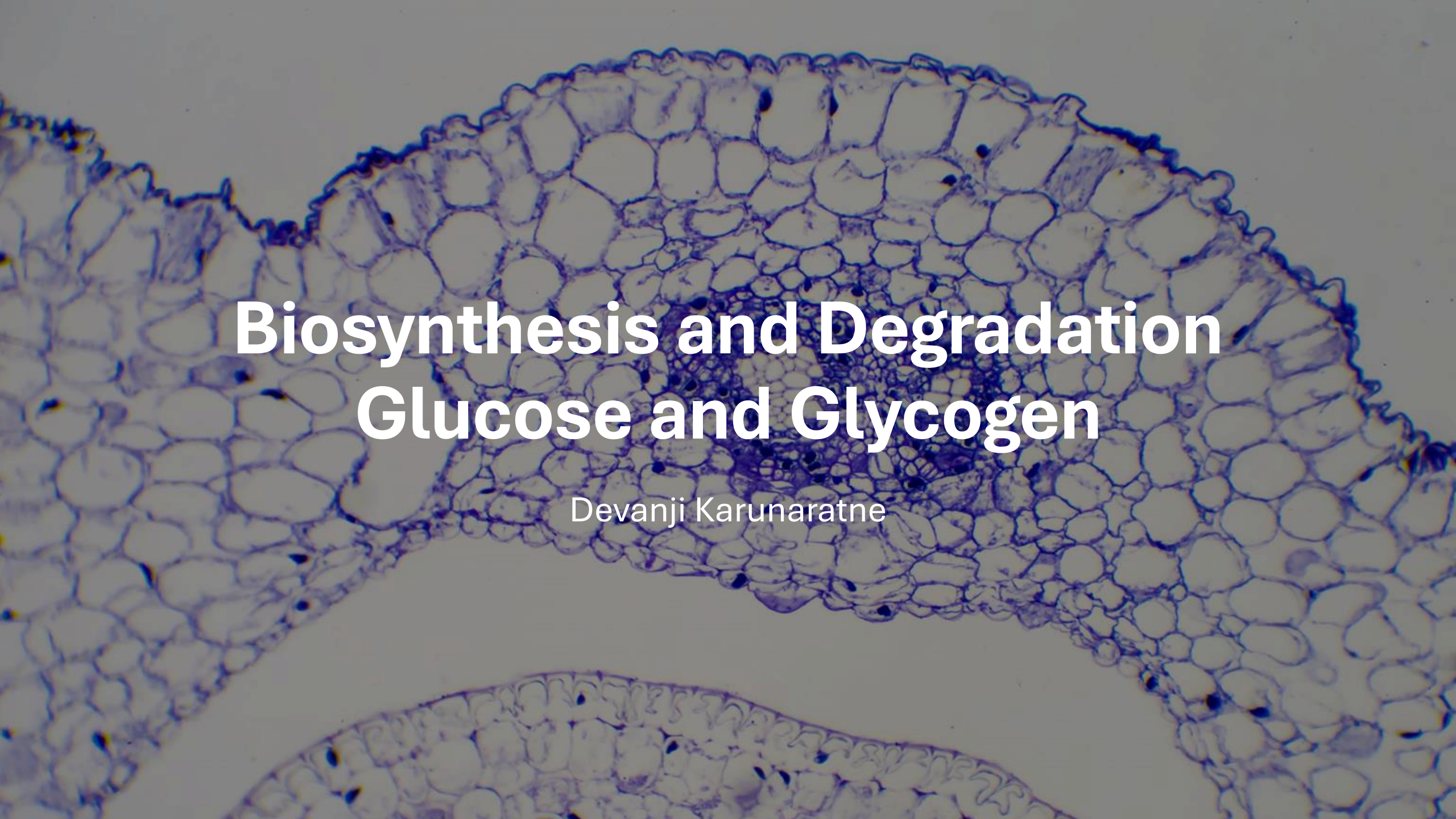


A microscopic image of a plant tissue section, likely a leaf cross-section, stained with a blue dye. The image shows a dense network of cells with prominent cell walls. The cells are arranged in a somewhat regular pattern, with some larger cells and some smaller ones. The overall structure is complex and detailed.

Biosynthesis and Degradation Glucose and Glycogen

Devanji Karunaratne

Central Role of Glucose in Energy Homeostasis

- Our body's ability to maintain a stable blood glucose concentration is critical.
- Certain tissues, like the brain and red blood cells, are almost entirely dependent on glucose for their energy needs.
- **Break down glucose** for immediate energy (Glycolysis)
- **Synthesize new glucose** from non-carbohydrate precursors (Gluconeogenesis)
- **Store glucose** as a readily mobilizable polymer, glycogen (Glycogenesis)
- **Release glucose** from its stored form, glycogen (Glycogenolysis)

Glucose Degradation: Glycolysis

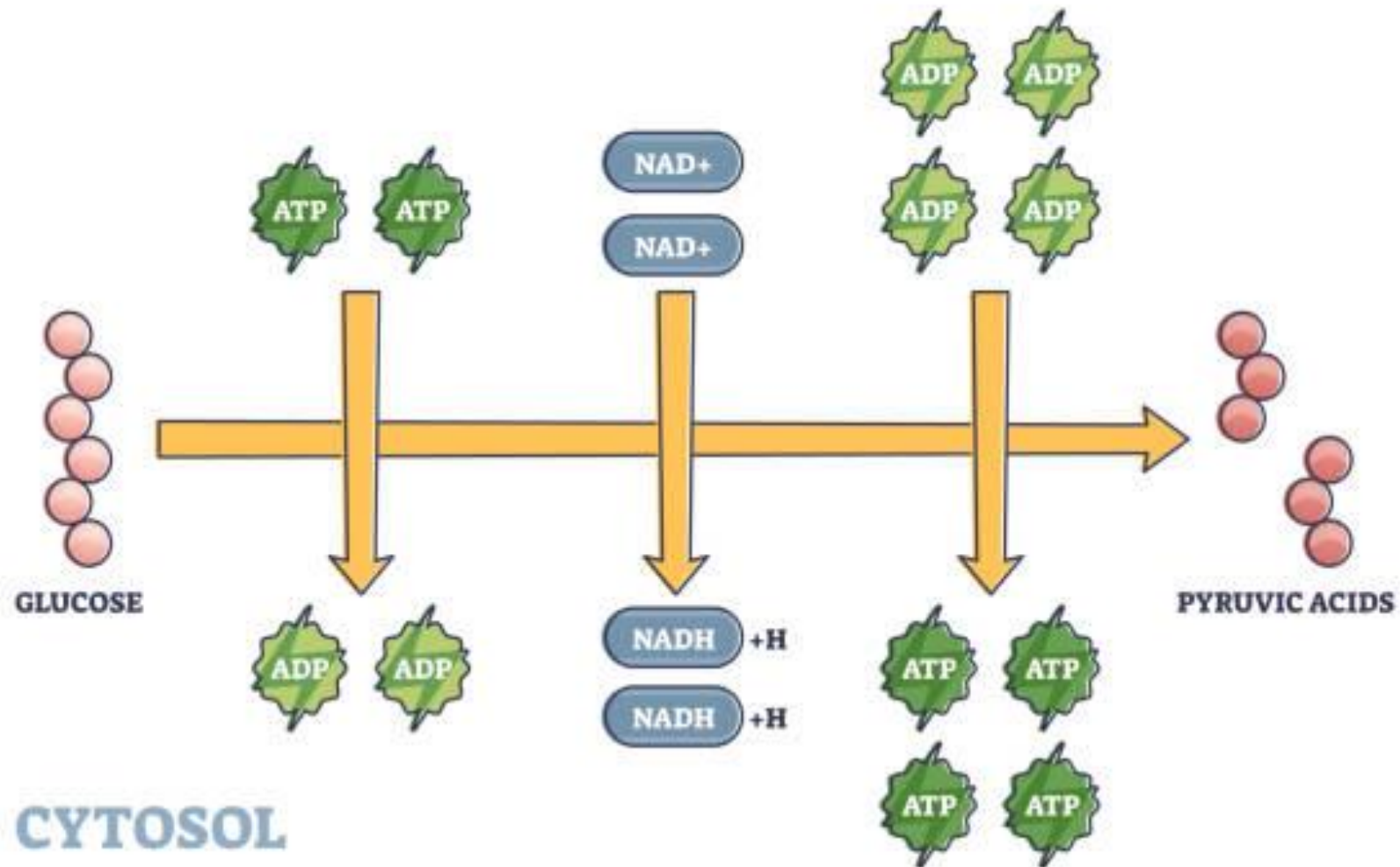
Glucose Degradation: Glycolysis

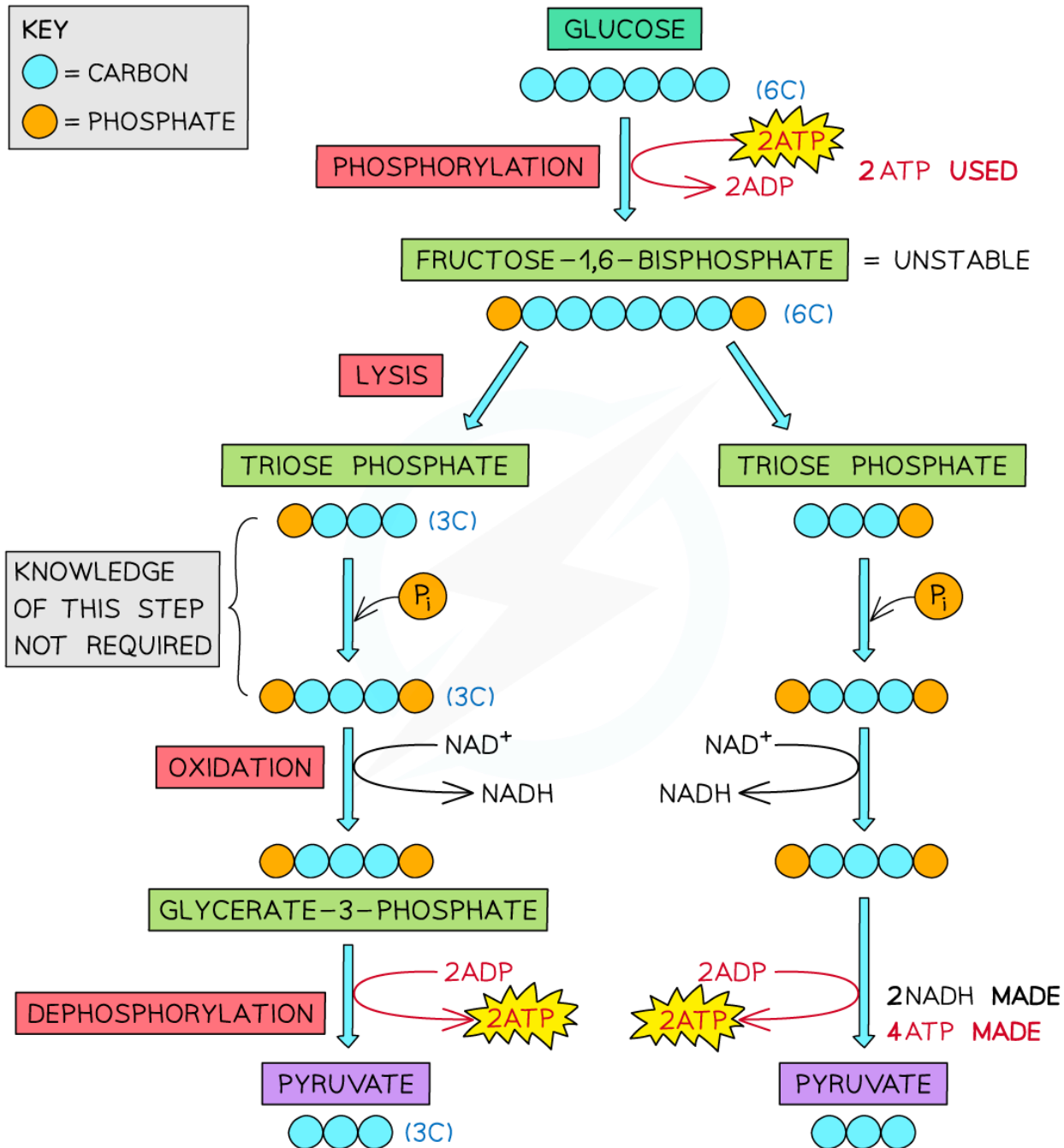
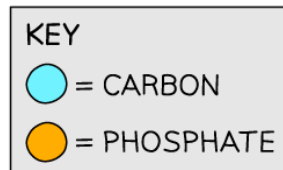
- **Glycolysis** is the universal pathway for glucose catabolism, literally meaning "splitting of sugar."
- **Purpose:** To extract energy from glucose by converting one molecule of 6-carbon glucose into two molecules of 3-carbon **pyruvate**, generating ATP (adenosine triphosphate) and NADH (nicotinamide adenine dinucleotide) in the process. This is the first stage of both aerobic and anaerobic respiration.
- **Location:** Occurs entirely in the **cytosol** of all cells.
- **Overall Reaction:**



- **Two Phases:** Glycolysis proceeds through 10 distinct enzymatic steps, divided into two phases:

GLYCOLYSIS





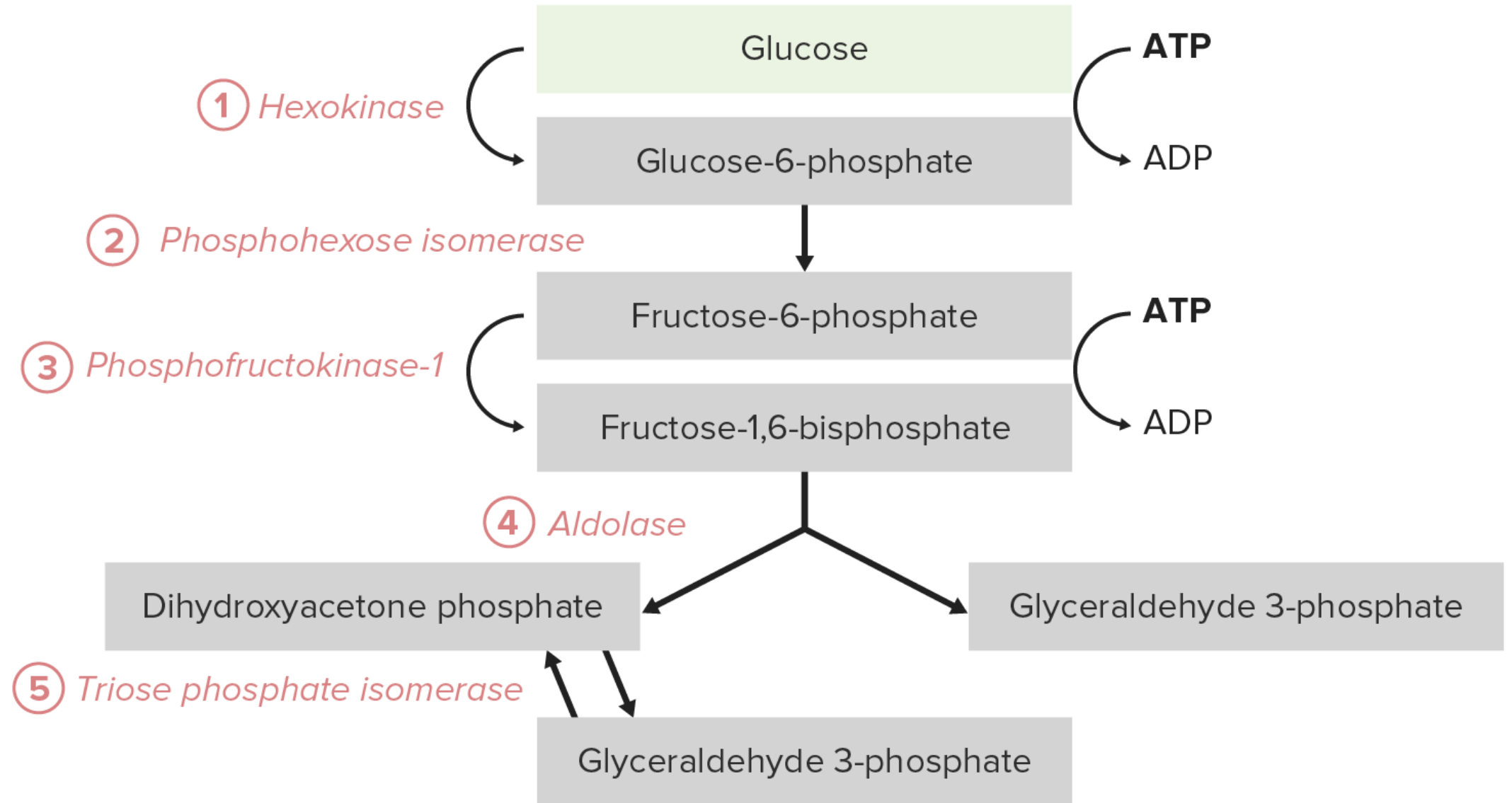
Energy Investment Phase

This phase consumes 2 molecules of ATP to convert glucose into two molecules of glyceraldehyde-3-phosphate.

It "primes" the glucose molecule for cleavage and traps it within the cell.

1. **Phosphorylation of Glucose**
2. **Isomerization**
3. **Second Phosphorylation**
4. **Cleavage**
5. **Isomerization**

1st half of glycolysis

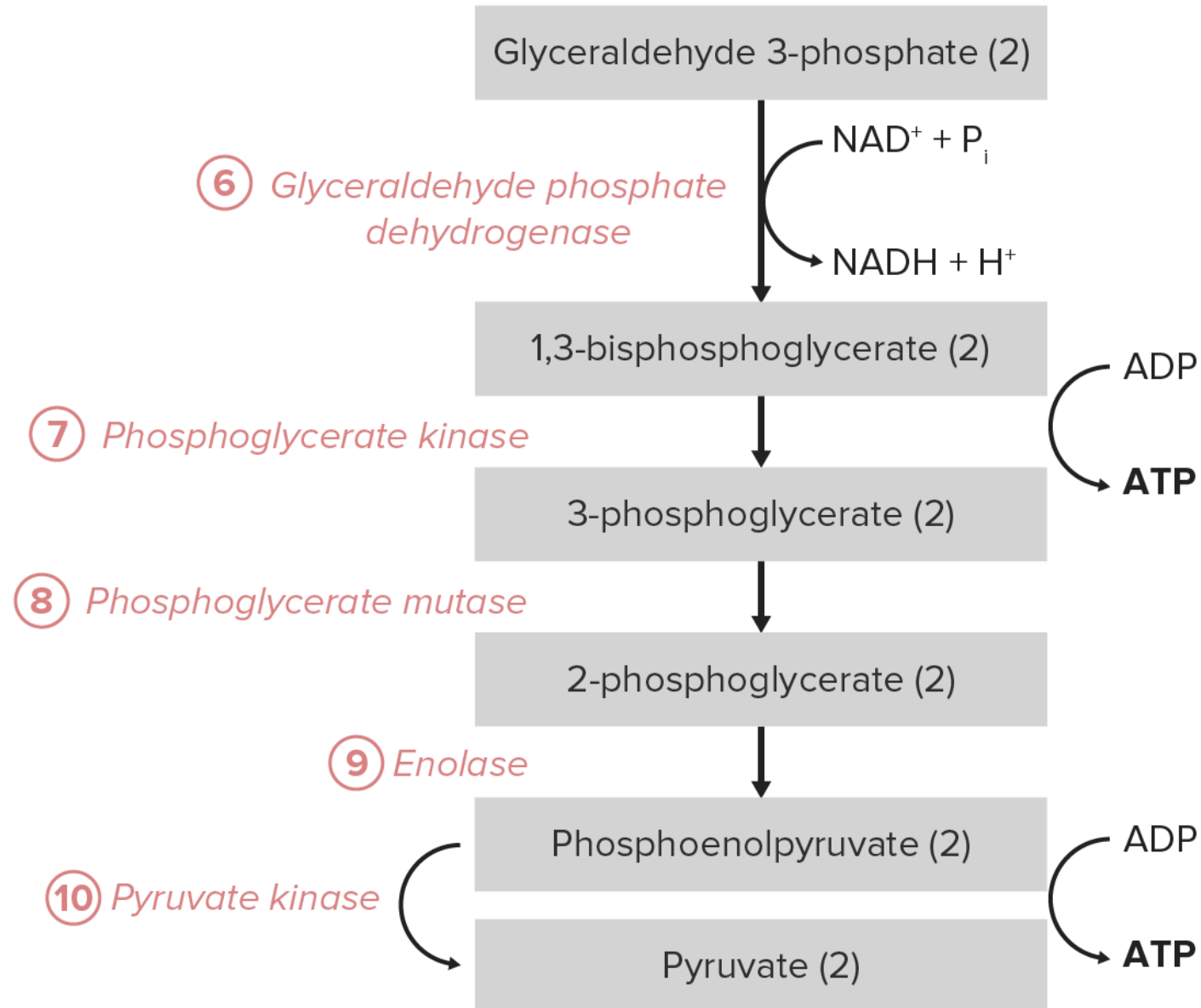


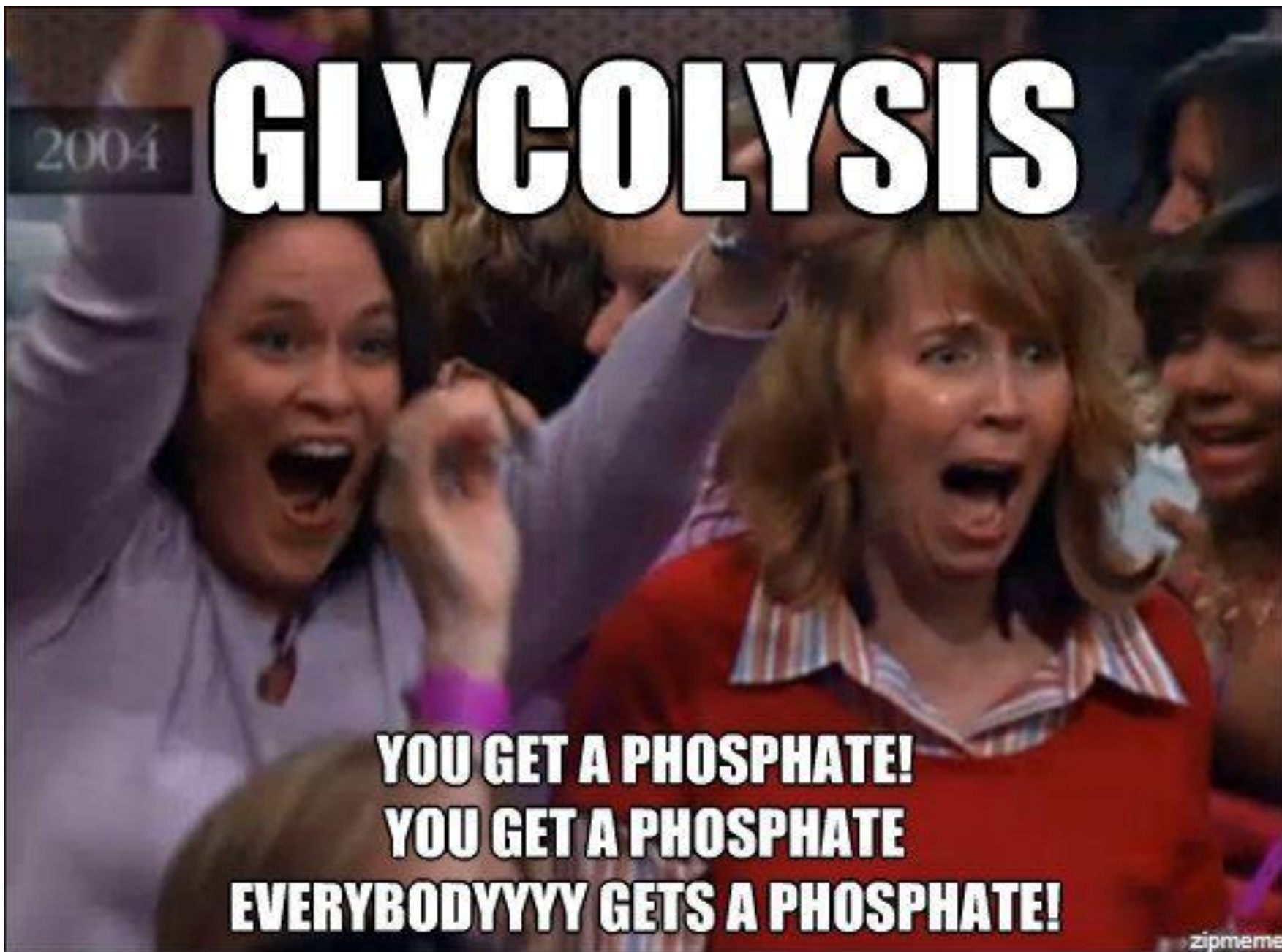
Energy Payoff Phase

This phase generates 4 molecules of ATP (net 2 ATP) and 2 molecules of NADH from the two GAP molecules.

1. Oxidation and Phosphorylation
2. First Substrate-Level Phosphorylation
3. Phosphate Group Transfer (Isomerization)
4. Dehydration
5. Second Substrate-Level Phosphorylation

2nd half of glycolysis





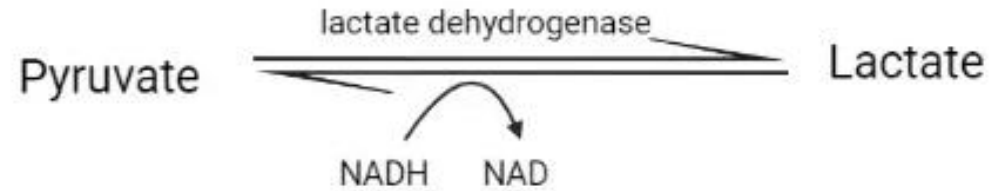
Fate of Pyruvate

- **Aerobic Conditions (Presence of O₂):** Pyruvate is transported into the mitochondria and converted to **Acetyl-CoA** (by pyruvate dehydrogenase complex). Acetyl-CoA then enters the **Citric Acid Cycle (Krebs Cycle)** and **Oxidative Phosphorylation** (via the Electron Transport Chain), leading to the generation of a large amount of ATP (~30-32 ATP per glucose). This is the primary mode of energy production in most animal cells.
- **Anaerobic Conditions (Absence of O₂):** When oxygen is limited (e.g., in vigorously contracting muscle cells, or in red blood cells that lack mitochondria), pyruvate is converted to **Lactate** (in animals) or **Ethanol and CO₂** (in yeast). This process, called **fermentation**, regenerates NAD⁺ from NADH, which is essential to keep glycolysis running and produce a small amount of ATP (2 ATP per glucose) in the absence of oxygen.

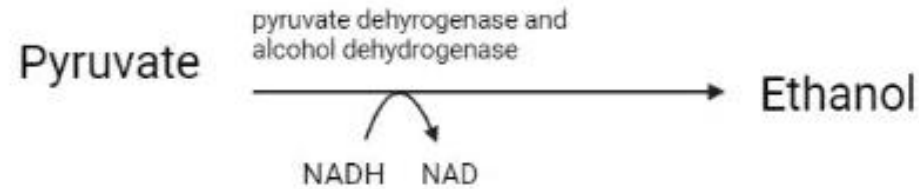
FATE OF PYRUVATE

Anaerobic condition

Lactic Acid Fermentation

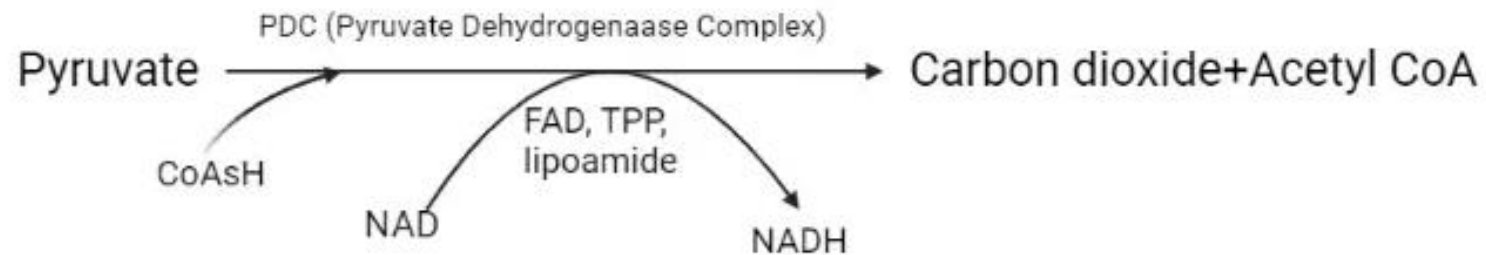


Ethanol Fermentation



Aerobic condition

Acetyl CoA formation



**IF YOU START STUDYING GLYCOLYSIS
3 HRS BEFORE THE BIOCHEM EXAM**

